Distributed FLISR Algorithm for Smart Grid Self-Reconfiguration based on IEC61850

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***Abstract*—This paper proposes a distributed algorithm for Fault Location, Isolation, and Service Restoration (FLISR) to be implemented as part of the intelligent software for controlling the operation of each circuit breaker. The paper proposes an event driven finite-state machine design that assures the self-reconfiguration of the distribution power grid when a fault occurs on a certain section of a feeder. The benefits of such an approach are first of all the modularity, as each breaker is controlled by the same algorithm and requires data exchange only with the neighboring protection devices. Secondly, by using the event driven technique, the FLISR algorithm is perfectly suited for taking advantage of the features and benefits defined by the IEC 61850 protocol. Using GOOSE messages for transferring status information between the neighboring circuit breakers, the proposed distributed algorithms act in a collective manner for reconfiguring the distribution grid.**

Keywords—FLISR, distributed algorithm, finite-state machine, IEC 61850, self-reconfiguration, Smart Grid.

I. INTRODUCTION

The extensive and continuous change of the power grid at all the levels is raising new technical challenges that must be coped with in order to insure the operation of the future power system.

As the future power grids extend their complexity due to the penetration of distributed energy resources, it is required to apply interdisciplinary technologies and techniques inspired from a variety of research fields far beyond the traditional power electronics [1]. The new techniques to introduce benefits in smart grids include, among others, game theory, artificial intelligence, and agent based approaches.

The two main challenges of future power grids can be defined in two major groups [2]: to provide optimal operation conditions for all the electrical devices of the system and to be able to cope with faults that are interrupting the operation of the power system.

The penetration of renewable sources in the distribution grid and at the consumer level is producing a change in the power flow direction, a change that is forcing the grid to take measures for assuring the stable operation of the system. Along with the power and energy management issues, the grid must implement solutions for restoring power in the case of a fault [3].

Fault location, isolation, and system restoration (FLISR) is an important issue for the future grid. This allows an automated isolation of a faulted section of a feeder and the reconfiguration of the grid in order to maximize the power restoration [2-5].

Currently, the FLISR algorithms are implemented both in centralized and distributed approach, each having its own advantages and disadvantages [3]. For example, the centralized algorithm computes the global optimal reconfiguration of the grid, considering all the constraints and using measurements and status information for all across the grid. On the other hand, the decentralized approach has the advantage of computation speed as it computes a local optimal solution for the grid reconfiguration.

For assessing a FLISR algorithm key performance indexes are defined [2] as: system average interruption duration index (SAIDI), customers' average restoration time index (CARTI), and fraction of served nodes (FoS).

This paper investigates the distributed FLISR algorithms and emphasizes on another advantage of this approach: the modularity and the short time for configuring the necessary circuit breakers when new sections are defined in the grid.

The paper focuses on the design, implementation, and validation of a new FLISR algorithm. The FoS is considered as indicator for the algorithm's efficiency. The developed algorithm is designed to use the IEC61850 standard protocol [6-8] and the defined GOOSE messages for sending status information between the breakers.

II. PROPOSED FLISR ALGORITHM

For the proposed distributed FLISR algorithm, the starting premise is that each circuit breaker is equipped with a computational module on which the algorithm is implemented and an Ethernet communication module for exchanging the IEC 61850 messages. The algorithm is designed to perform in meshed grids with radial feeder configurations specific to the medium and low voltage distribution grids and is implemented as a finite-state machine. The event driven definition of state transitions are perfectly suited for using the GOOSE message definition from the IEC61850. In addition, the IEC61850 standard protocol defines the time delay of exchanging GOOSE messages between two devices to be 5 ms [8].

The different instances of the same distributed FLISR algorithm running on each of the breakers are acting on the whole system as the algorithm searches for a grid configuration from the set of all possible configurations of the power system in order to provide power to a maximum number of loads. The service restoration is a global objective [7-10] for which the present work proposes a distributed solution, implemented on each device.

One mention has to be made: the approach presented in this work does not guaranty the optimal solution for reconfiguring the power network. This optimum can only be achieved using a holistic view over the system, from a centralized control that has access to information regarding the state of the entire system. As the reconfiguration is NP complete [2,11], all the possible configurations must be explored and compared in order to select the best solution.

The current work proposes one distributed algorithm for each of the three types of circuit breakers devices found on a distribution power grid:

-the circuit breaker – used to connect the feeder to the transformer

- the sectionalizing breaker – used to separate different sections on the same feeder

- the tie breaker – used for interconnecting different feeders

In other words, for each of the three types of breakers a different agent is defined with specific functions. The agent based approach can be referenced in the literature as it is described in [2,12,13]. In addition, in this proposed work, each instance of the FLISR algorithm exchanges information between adjacent players (breakers) for arriving to a global solution that will solve the reconfiguration problem and improve the SAIDI and FoS indexes. These information exchanges can happen multiple times as the algorithm searches for a solution.

The proposed algorithm can be analyzed from two perspectives: firstly, the local level, when running on the computational module of each breaker, the algorithm uses the specific states defined to operate the defined functions. Secondly, the global perspective, where the interactions between the multiple instances of the same distributed algorithm are providing a global evolution.

A. The global perspective of the FLISR algorithm

Seen from the global perspective, this approach is related to general game theory applications where multiple players with the same behavior are shaping the global evolution of the system [14-17]. Each breaker is using the same algorithm (or strategy for continuing the parallel to the game theory [18]) when interacting to the other breakers for defining the operation of the entire system.

Figure 1 presents the feeder configuration used for testing the proposed algorithm. There are eight sections defined on this feeder, each powering a certain load. There are also two tie breakers connecting the feeder to adjacent feeders for providing power in case of necessity. Details on the values used in the simulation are provided in Section III. In addition, Fig. 1 presents the breaker's status after the reconfiguration of the simulated scenario.

By having all the breakers acting in a collective manner, the overall algorithm seen from the global perspective, presents the following steps:

- when a fault short circuit occurs, this is detected by the algorithm running on the breaker, opens the circuit and sends two GOOSE messages to the next, downstream breaker: a trip message and the current value measured by the breaker prior to the fault; this second breaker acts on receiving the trip message from the upstream neighbor, opens its circuit, and sends the same two messages to the next downstream breaker. The process is repeated until the last breaker on the feeder is opened; this is represented by *Phase I*, in Fig.1.

- on the second phase, *Phase II*, of the FLISR algorithm, the feeder is covered upstream starting from the last breaker up, in order to find a tie breaker able to provide power to sections on the faulted feeder from a neighboring feeder

- when a tie breaker connected to a feeder with available current capacity is found, it is closed and the corresponding section of the feeder is energized. This is the beginning of *Phase III* as represented in Fig. 1.

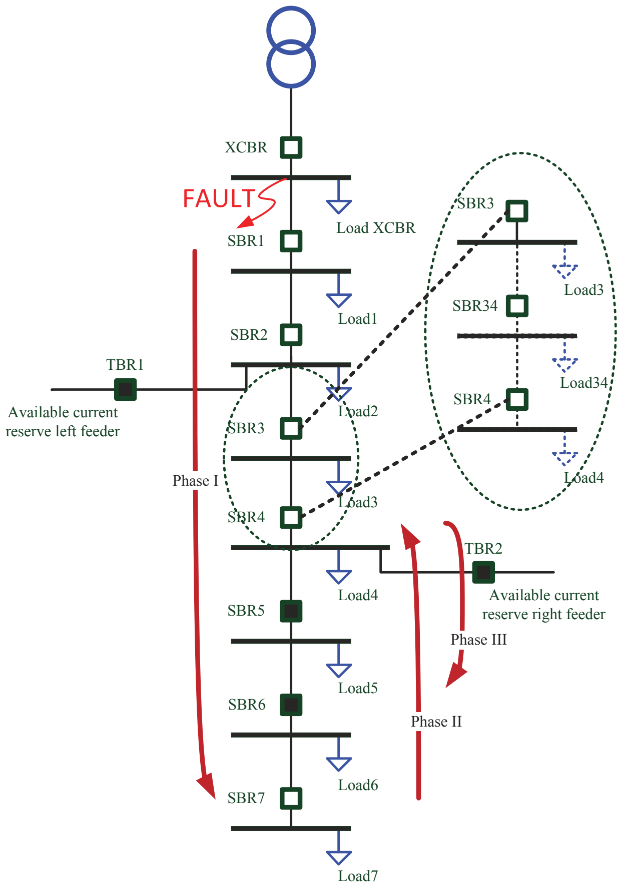

Figure 1.The feeder configuration for testing the algorithm. (breaker's status after the reconfiguration of the simulated scenario).

- *Phase III* is finalized when the current capacity of the neighboring feeder providing the current for the faulted feeder is reached or all the downstream sections are energized.
- at this moment, a new upstream cover of the feeder is initiated in order to connect sections situated upstream from the tie breaker or, in the case the current capacity is reached, to find another tie breaker for providing current to the de-energized sections. This phase is similar to Phase II, having in addition the objective of connecting sections of the feeder to an energized downstream section.

In the case of a fault, the algorithm covers the feeder from the breaker sensing the fault to the last breaker on the feeder and then searches upstream for a source of available current (a tie breaker connecting an adjacent feeder). The reason is that, in this way, the algorithm is able to calculate the load on each section and to supply this information to the breaker linked to the corresponding section, as it is detailed in *SectionII.B*.

The major benefit of this approach is the fact that this reduces the effort and the time of reconfiguring the protection devices in case that new sections are being defined on the feeder as shown in Fig. 1 (e.g. when introducing a new breaker SBR34 for serving Load34).Compared with a centralized algorithm that requires knowledge of the entire structure of the system, in the proposed algorithm, only the neighboring breakers need to exchange information about their status. In case that a new section is introduced, only two breakers (in this example SBR3 and SBR4) need to have the IP and MAC addresses modified to refer to the new breaker (SBR34) according to the IEC61850 definitions. This process is similar to a well-known technique in computer science, which is the chain list [18].

B. The local perspective of the FLISR algorithm

Until this section, the paper presented the evolution of the algorithm perceived from the global perspective, the combined effect of using multiple instances of the distributed FLISR algorithm acting for reconfiguring the system. In this section the focus is set on presenting the distributed algorithm that is implemented on each breaker.

As mentioned before, the algorithm is developed as a finite-state machine, with states defined for different stages of operation and transitions. These transitions are conditioned by the signals received from the local measurements or from the neighboring breakers. Fig. 2 provides a mapping of the states defined by the algorithm. The states have been grouped in larger states, numbered from 1 to 7, in order to provide a clearer overview on the algorithm.

The initial state (***State1***) is active until an overcurrent is detected from the local current measurements or a *trip*/*primary trip* message is received from the upstream breaker.

When the algorithm running on a breaker identifies an overcurrent on its section of the feeder, it will initiate a transition to a state consisting in opening the circuit and sending the neighboring downstream breaker a dataset of GOOSE messages containing a trip and the load current measured at the breaker's level prior to the fault (***State2***). This current represents the load of the entire downstream feeder and will be used by the neighboring downstream breaker to calculate the load on the current section. As the fault needs to remain isolated from the rest of the system after the reconfiguration the algorithm sets a flag signaling this breaker being one end of the faulty section. This breaker will remain open until a technical crew will eliminate the problem that produced the fault and reset the breaker.

State3 is responsible for the first cover of the feeder from the breaker sensing the fault down to the last breaker on the feeder. The first downstream breaker from the location of the fault will receive a *primary trip* message in order for the algorithm to identify the second breaker that frames the faulty section of the feeder. In this way, when a reconfiguration is searched, these two breakers will always remain in the open position. When a *Trip* message is received, a transition is made to a state designed to calculate the load on the upstream section. This part corresponds to *Phase I* of the general perspective on the algorithm as presented in Fig. 1 when the feeder is covered downstream.

In ***State4***, 4 different sub-states are defined for covering the feeder as mentioned in *Section II.A* and represented in Fig.1 as *PhaseII*. Basically, during this states, the algorithm will send to the upstream breaker two information: the status regarding the load on the upper section (for example, the algorithm on SBR4 will calculate the value of Load3) and the current measured at the breaker level. These information will indicate each breaker the status of the neighboring sections.

State5, as shown in Fig. 2, contains states defined for connecting tie breakers and energizing one section at a time. The tie breaker will receive a GOOSE message from the algorithm containing the necessary power for connecting the downstream section of a breaker, which is the local load (LoadLocalSection) and will close the circuit if this power (or current capacity) can be provided.

State6 implements the action corresponding to *PhaseIII* in Fig. 1, of connecting downstream sections until the end of the feeder or until the current capacity of the feeder providing support for the faulty region is reached.

State7 is defined as an idle state, when the algorithm does not take any action. After an intervention crew arrives and the fault is cleared, the breaker is reset and the finite-state machine goes back to *State1*.

Table I provides the list of signals used by the FLISR algorithm developed for the sectioning breakers. These signals will be mapped to new proposed LogicalNodes according to the IEC 61850 protocol and GOOSE messages will be used to exchange this information.

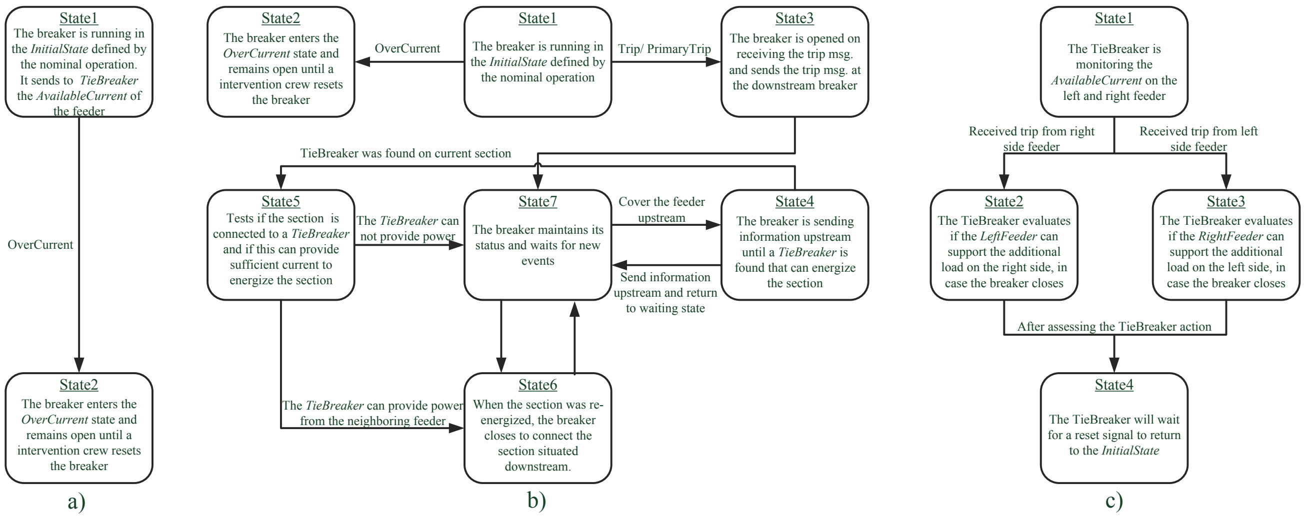

Figure 2.The distributed FLISR algorithm developed for a) a circuit breaker, b) a sectionalizing breaker, and c) a tie breaker, designed as a finite-state machine.

TABLE I. THE VALUES REQUIREDBY THE FLISR ALGORITHM

Inputs			**Outputs**		
Name	**Type**	**Description**	**Name**	**Type**	**Description**
Initialization parameters					
LastOnFeeder	Bool	Is *True* if the current breaker is the last on feeder			
TieSwitch	Bool	Is *True* if the breaker is a Tie Breaker			
InitVal	Bool	Sets the initial state of breaker			
Signals used at the breaker level (not used for communication)					
Current	Float	The measured current	Close	Bool	Provides the status of the breaker
Signals used to exchange trip status information					
OverCurrent	Bool	Is *True* is an OverCurrent is detected			
			SendTripTS	Bool	Sent trip message to the tie breaker
RecTrip	Bool	Received message signaling the trip of a upstream breaker	SendTrip	Bool	Sent trip message to the downstream breaker
RecTripPrimary	Bool	Received message signaling the fault occurred upstream of the current breaker	SendPrimaryTrip	Bool	Sent trip message to the breaker downstream the fault
Signals used to exchange other status information					
RecNSLoad	Float	Received current value from the upstream breaker	SendNSLoad	Float	Sent value of the local measured current to the downstream breaker
RecSNLoadN	Float	Received value of the calculated current section load from the downstream breaker	SendSNLoadN	Float	Sent value of the calculated upstream's section load to the upstream breaker
RecSNLoadLocalSection	Float	Received value of the section's load	SendSNLoadLocalSetion	Float	Sent the value of the section's load
RecLoadLocalSWS	Float	Received current value from the upstream breaker after reconfiguration started	SendLoadLocalSWS	Float	Sent value of the local measured current to the downstream breaker after reconfiguration started
Signals exchanged with the tie breaker					
TieOpen	Bool	Signals the opening of a Tie breaker	SendTie	Bool	
			SendTieLoad	Float	Sent the tie breaker the value of the load
Signals concerning the cover direction (upstream or downstream the feeder)					
RunN	Bool	Received signals for the coverage of the feeder upstream	SendNRunN	Bool	Sent signal for the upstream breaker that the algorithm is covering the feeder upstream
RunS	Bool	Received signals for the coverage of the feeder downstream	SendSRunS	bool	Sent signal for the upstream breaker that the algorithm is covering the feeder downstream

III. Simulation and Results

For validation, the distribution system presented in Fig. 1 was modeled in Matlab/Simulink using the SimPowerSystems toolbox. The proposed algorithm was designed as a finite-state machine implemented with the StateFlow toolbox. The study case presents the operation of the FLISR algorithm when a fault occurs at the level of the section between XCBR and SBR1, as presented in the initial Fig. 1. The current values for the considered simulation are presented in Table II. In the same table are shown the magnitude of the resistive loads and the RMS values of the corresponding load currents, before the fault.

Initially, TBR1 and TBR2 are *opened* and all the breakers on the feeder (SBR1 to SBR7) are *closed*. The reserve from the left feeder is 60A and from the right the value is 100A.

As the fault occurs and is detected by the XCBR breaker, the FLISR algorithm is activated. The evolution of the TRIP status of each breaker involved in the reconfiguration of the system is illustrated in Fig. 3. This evolution is correlated with the *Phases* presented in Fig. 1 and discussed in Section II. After the fault detection at t=2s, all the sectionalizing breakers are tripping (opening the circuit) as according to *PhaseI*. After the last breaker on the feeder has been opened, *PhaseII* is started – responsible to find a tie breaker that connects the faulted feeder to another, neighboring feeder that provides available current capacity. In Fig. 3,*PhaseII* evolution is visible as the first breaker to close is the tie breaker TBR2, connecting the faulty feeder to the feeder on the right that has 100A current capacity available. As presented in the algorithm's description, after finding and closing a tie breaker, status messages are being exchanged between the sectionalizing breakers running the distributed FLISR algorithm in order to energize other sections as well. Fig. 3 and Fig. 4 present this phase of the algorithm as the next breakers to be closed are SBR5 and SBR6, energizing Load5 and Load6.The last to close is TBR1 energizing Load2 (this corresponds to Fig.1).

TABLE II. Simulation Settings and measurements

Sectioning breaker	Current [A]	Load name	Load value [MW]	RMS Load current[A]
XCBR	387.6	LoadXCBR	3.0	86.5
SBR1	301.5	Load1	3.0	86.5
SBR2	215.3	Load2	1.8	51.9
SBR3	163.6	Load3	2.0	57.66
SBR4	106.1	Load4	1.2	34.6
SBR5	71.6	Load5	1.0	28.83
SBR6	42.9	Load6	0.7	20.18
SBR7	22.8	Load7	0.8	23.06

Figure 4 represents the RMS values of the load currents for loads 2 to 6. This evolution of the current is correlated to the tripping of the breakers (Fig.3) and clearly shows the sequence of breakers opening and closing. This reconfiguration sequence is the result of the proposed distributed FLISR algorithm described in Section II.

In Table II, the difference between the RMS Load current measured at the load side and the current value measured as the difference between two adjacent breakers after the fault (for example Load1 and SBR1-SBR2) is due to the measuring method used to acquire the past value of the current, prior to the fault. As a filter was used to average the past values, due to the delay between the moment the fault occurs and the moment it is detected, the averaged value of the current is slightly decreased from the one of the real current prior to the fault.

The time needed for the algorithm to reconfigure the system has a qualitative purpose, for a more accurate estimation, a real-time simulation is needed. However, as the algorithm is based on GOOSE messages that use 5ms for transmission over the Ethernet, this reconfiguration time is expected to be close to the one obtained in the simulation, or even better, as shown in the preliminary test in Fig. 6.

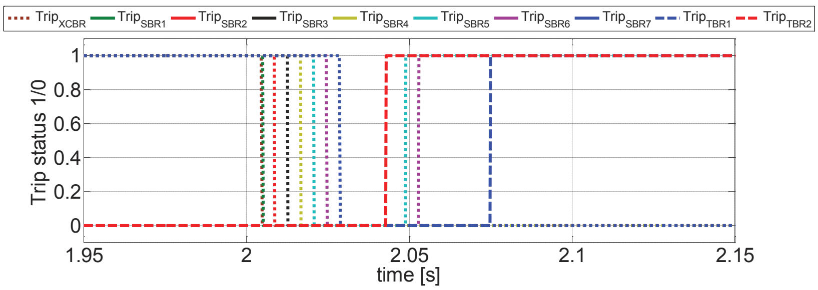

Figure 3.The sequence of TRIP status for all the breakers in the system.

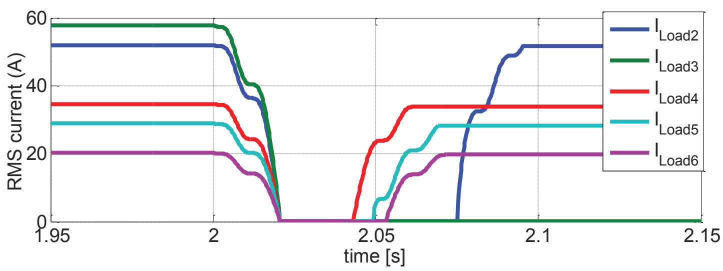

Figure 4.Current load values before and after the FLISR algorithm reconfiguration.

In the next stage of the work, one breaker will be implemented on a processor based development board (DK60) capable of IEC61850 communication. The board will interact with the real-time simulator (OPAL-RT target) for testing the operation of the algorithm when using GOOSE message exchange between different devices over the Ethernet. This will further validate the algorithm as it will run in real-time and, in this way, the key performance indexes defined at the beginning of the paper can be evaluated. The hardware used for testing the algorithm in real-time simulations is presented in Fig. 5.

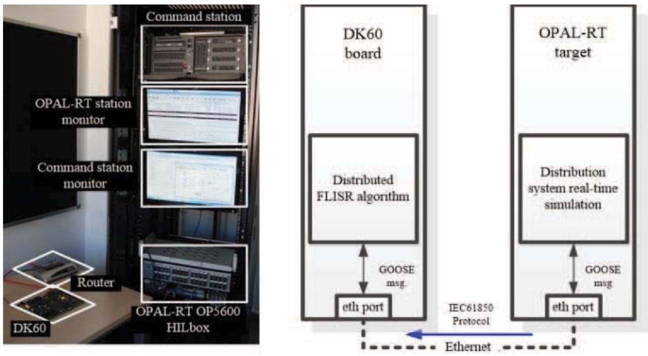

Figure 5.The available hardware for real-time validation of the algorithm.

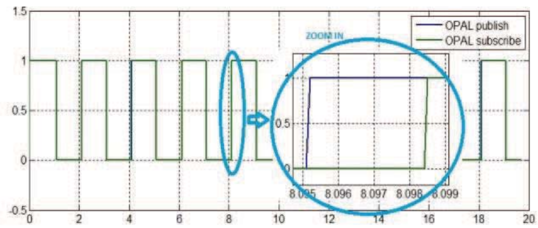

Figure 6.Time response for a GOOSE message in HIL simulation.

IV. CONCLUSIONS

In this paper a distributed Fault Location, Isolation, and Service Restoration algorithm implemented on each sectionalizing breaker has been proposed and tested. Using this algorithm, the breakers provide the overall system the self-reconfiguration ability.

The proposed FLISR algorithm offers great benefits when physical changes are made to the grid configuration: for example, defining a new section and installing a new breaker will be made with very few changes, as each breaker is exchanging information only with the neighboring breakers. In this way, the time for changing the configuration of the protection devices of the grid is significantly reduced.

The proposed algorithm is suitable also for cases when distributed energy resources are being used at the grid level as the distributed algorithm exchanges the status of the section load which can be both positive and negative, according to the power balance between the local consumption and generation at the bus level.

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